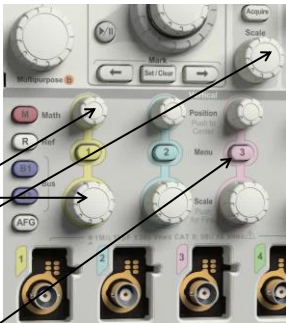


Basic settings



Voltage division: Volts/division

Vertical shift: (vertical pos.)

Time base: Seconds/division

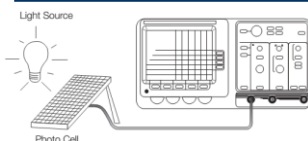
Input setup
DC/AC coupling; in case of GND – AC coupling, a capacitor is connected in series with the input, so the input sees only the alternating signal, the direct voltage (DC) is filtered out. GND grounds the input.

Termination: 50 Ω, 75 Ω, 1 MΩ – setting input resistance

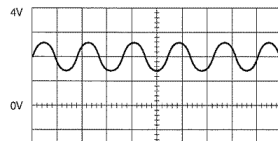
Invert ON/OFF – optional, displays the signal × (-1)

Bandwidth FULL/20MHz – the input bandwidth can be limited

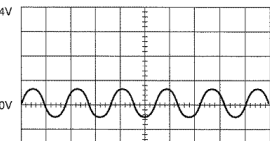
Experiments: measuring "blinking" light

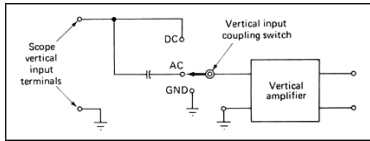


DC Coupling of a 1 V_{pp} Sine Wave with a 2 V DC Component



AC Coupling of the Same Signal





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Trigger settings

Without synchronization (triggering), the waveforms will appear in different phases on the screen

Source – which channel is used for triggering

Level – what voltage threshold to trigger to

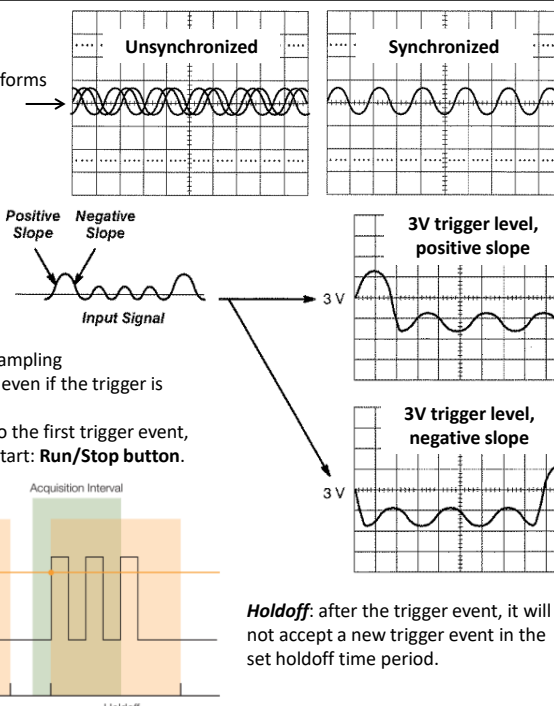
Slope – trigger to positive or negative slope

Coupling – trigger coupling (AC, DC, HF reject, LF reject, stb.)

Normal mode: records a curve after each trigger event

Auto mode: if it does not get a trigger signal, sampling is started automatically ("let us see something even if the trigger is uncertain")

Single shot: records the curve corresponding to the first trigger event, then stops (does not wait for more curves) Restart: **Run/Stop button**.

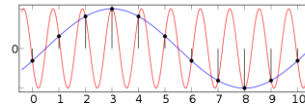


Holdoff: after the trigger event, it will not accept a new trigger event in the set holdoff time period.

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Acquire menu

Experiment: aliasing



Mode:

Sample – the raw, sampled datapoints are displayed

Enhanced resolution – More neighboring datapoints are averaged together, thus the vertical voltage resolution is improved, but the horizontal, time resolution is decreased. Repeating signals are not needed for this.

Average – averages consecutive waveforms, thus the voltage resolution is enhanced, and time-resolution is not decreased, however, longer waiting period is needed and can be used only with precisely repeating signals.

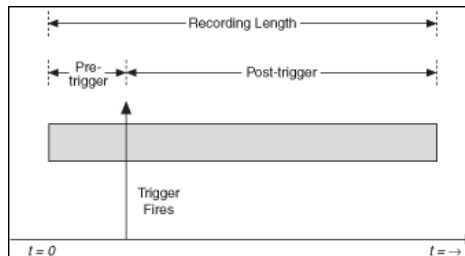
Etc.

Record length: pl. 1k, 10k, 100k, 1M, 5M, 10M points – how many points should a sampling take. For example: we want to resolve one period of a 50 Hz signal, so the sampling time-window is 20 ms. In case of 10M datapoints, the time-resolutions is 2 ns, thus, superimposed to the 50 Hz signal, even a few hundred MHz signal can be resolved. Processing a lot of datapoints can take long, especially if we do additional mathematical operations.



XY mode: The horizontal axis does not stand for time, rather we display the measured voltage as a function of the voltage of another channel. E.g. Lissajous curves

Trigger position, pretrigger: where the sampling window should be with respect to the trigger signal. We can record datapoints before the trigger (pretrigger)!



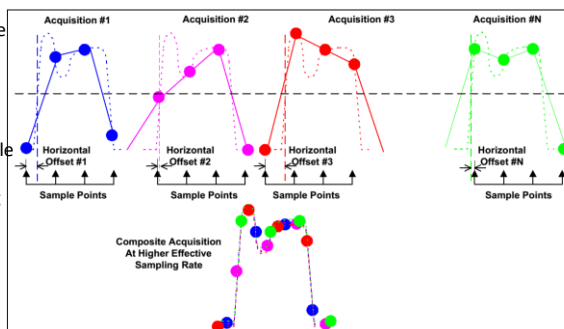
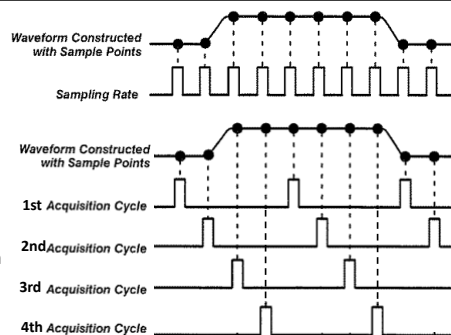
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Sampling methods

Real time sampling: measurement points are taken at intervals corresponding to the clock signal. Even a single curve can be taken, no repetitive signal is needed.

Sequential equivalent time sampling: after a trigger event, a measurement point is taken with time delay Δt , and for successive trigger events, this Δt is always increased a little. Advantage: significantly higher time resolution can be achieved than with real time sampling. Disadvantages: can only be used for repetitive signals with a precise trigger event and no possibility to measure before the trigger event (pretrigger).

Random equivalent time sampling: the scope samples continuously, with trigger events occurring at random times relative to the sampling period (the measured signal and the clock signal are not synchronous). The time between the trigger and the first sampling after the trigger is measured much more accurately than the sampling period. Measuring the repetitive waveform at multiple trigger events and shifting the waveforms together with the measured trigger-sampling event delays the waveform can be resolved with a much better temporal resolution than with real time sampling. Pretriggering is also possible, but this method is slower than sequential equivalent time sampling.



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Measurements, math, cursors

Measurements: numerous measurements can be done with the recorded waveform, e.g. *period, frequency, mean voltage, RMS voltage, peak-to-peak voltage, maximum/minimum voltage, rise time/fall time*, etc.

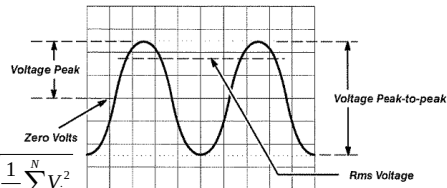
Definition of RMS voltage for continuous and discrete signal:

$$V_{\text{RMS}} = \sqrt{\frac{1}{T} \int_0^T [V(t)]^2 dt}; \quad V_{\text{RMS}} = \sqrt{\frac{1}{N} \sum_{i=1}^N V_i^2}$$

Illustratively: the DC voltage, whose power is the same as the average power of the alternating signal

Measurements: with the recorded waveform, or waveforms, mathematical operations can be done, e.g. *addition, multiplication, numerical differentiation, Fourier transformation*, etc.

Cursors: a horizontal and a vertical cursor pair can be placed on the screen, whose position, and the distance between them can be read.



Waveform generator (AWG, arbitrary waveform generator)

D/A converter with accurate clock signal, with that *sine, sawtooth and square signs, pulses*, or even *arbitrary* programmed waveforms can be released. Frequency, amplitude, offset, symmetry, pulse width etc. can be set.

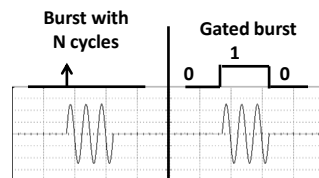
Advanced generators can be synchronized to external trigger signal (**burst mode**):

Burst with N cycles: gives out N periods from the signal after every trigger event

Gated burst: only gives out signal when trigger input is in state 1

Trigger input:

Generator output:



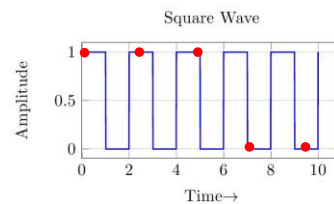
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Experiment: Measuring „blinking” light

We want to detect the light of a distant tungsten lamp. The detector shows a large DC signal, whose amplitude hardly changes as the light is switched on and off. How can we improve the measurement? As the light bulb is „blinking” with a frequency of 100 Hz (if the voltage alternates with 50 Hz, the intensity alternates with 100 Hz!), we can get rid of the DC background if we measure in AC coupling, and the signal can be amplified. Now the 100 Hz blinking shows, but very noisy, and it „crawls” horizontally on the screen. If we decrease the oscilloscope’s bandwidth, (e.g. from 100 MHz to 20 MHz), the signal becomes nicer, because the high-frequency noise is filtered out. The same can be achieved with the „enhanced resolution” function, in which the scope averages consecutive datapoints. With that, time resolution worsens, but voltage resolution improves. The horizontal „crawling” can be removed with appropriate triggering, however, we cannot trigger to a noisy signal, because the voltage noise will bring in noise in the triggering time as well. Best practice is to trigger to an external signal with large amplitude, that is in phase with our measured signal. (Most function generators can emit such a trigger signal, that is large-amplitude square sign in phase with the drive signal.) In our case we can trigger to the network (50 Hz) frequency, a function available on most oscilloscopes. If we succeeded with appropriately triggering, then averaging the consecutive curves (average function) will get rid of the noise on our 100 Hz signal.

Experiment: Aliasing

We see a large-amplitude, noiseless square signal on the scope, yet we cannot trigger to it, no matter how well we set the trigger level. What is the reason? In reality, we measure a square signal with frequency 100,010 Hz, in a 300 ms wide time window, with a sampling rate of 0.3 ms (1000samples/curve). Meanwhile, the period of our signal is ~0.01 ms, meaning we record a datapoint only in every 30th period, but because of the 10 Hz „off-tuning”, this point is recorded from a later part of the signal, and a latent square wave forms (aliasing). This problem can be avoided if the sampling rate is set in accordance with the period of the signal.



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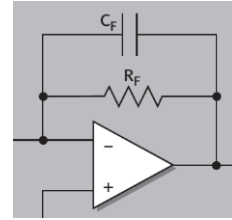
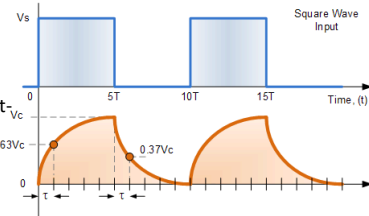
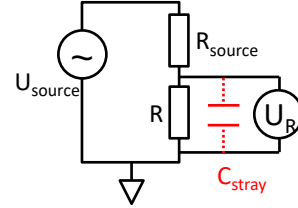
RC time constant

Cables and circuit elements always have a finite stray capacitance, which, together with the appropriate resistors in the circuit, forms an RC filter. Example: AC voltage source (U_{source}) is used to test resistance R in current generator mode. The circuit current is set with resistor $R_{\text{source}} \gg R$ (i.e. $I_{\text{source}} \approx U_{\text{source}}/R_{\text{source}}$), the voltage is measured at R . The resistor R and the long coax cable form a parallel RC circuit. Without the stray capacitance $U_R = I_{\text{source}} R$ voltage should drop on the resistor. However, if the absolute value of the stray capacitance's complex impedance becomes smaller than the resistance, $|Z_C| = 1/(\omega C_{\text{stray}}) < R$, most of the current flows through the capacitance and the measured U_R voltage strongly decreases. This means, that at frequency $2\pi f = \omega < 1/RC_{\text{stray}}$ the desired voltage is measured, while above the cut-off frequency $\omega = 1/RC_{\text{stray}}$ the measured voltage is cut-off, i.e. it becomes smaller as the frequency is increased.

The typical capacitance of a coax cable is 50-100pF/m, so with a 6m cable and $R = 3.3\text{k}\Omega$ (see experiment) we expect a cut-off at $f \sim 160\text{-}320\text{kHz}$.

If we drive the circuit with a square wave, we see a charge/discharge at the leading and trailing edges with a time constant of $\tau = RC_{\text{stray}}$

Another example: when measuring low currents, we usually use a current amplifier with a high feedback resistor. With $R_F = 1\text{G}\Omega$ feedback resistor and $C_F = 1\text{pF}$ stray capacitance, the instrument bandwidth is $f_{\text{max}} = 1/2\pi RC_F \sim 160\text{Hz}$, you cannot measure a signal with a higher frequency than this, i.e. you cannot measure small currents quickly. A significantly lower stray capacitance than this is not very feasible, but the stray capacitance can be somewhat compensated by clever tricks.



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Signal propagation in coaxial lines, telegrapher's equations

If the length of the cable is comparable to the wavelength of the electromagnetic wave propagating at a given frequency, then the description for low frequency circuits is no longer applicable!

Coaxial cable model: the outer (shield) wire is grounded. A dx long elementary segment of the inner conductor has a self-inductance coefficient $\tilde{L}dx$ and a resistance $\tilde{R}dx$. Furthermore, a dx long elementary segment of the inner conductor has a $\tilde{C}dx$ capacitance and $\tilde{G}dx$ conductance towards the shield. From these parameters the resistance of the inner wire and the conductance towards the shield is neglected ($\tilde{R} = 0, \tilde{G} = 0$). Due to the self-inductance, the voltage drop on the elementary section of wire dx is:

$$dU = -\tilde{L}dx \cdot \frac{dI}{dt},$$

while the variation of the current along a dx segment is determined by the charging and discharging of the capacitance towards the shield:

$$dI = -\tilde{C}dx \cdot \frac{dU}{dt}.$$

These can be used to write down the coupled differential equations on the current and voltage, the so-called telegrapher's equations:

$$\frac{\partial U(x,t)}{\partial x} = -\tilde{L} \cdot \frac{\partial I(x,t)}{\partial t}, \quad \frac{\partial I(x,t)}{\partial x} = -\tilde{C} \cdot \frac{\partial U(x,t)}{\partial t},$$

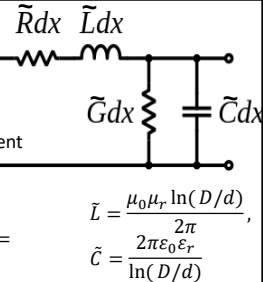
which simply give the following wave equations:

$$\frac{\partial^2 U(x,t)}{\partial x^2} = v_{\text{signal}}^2 \cdot \frac{\partial^2 U(x,t)}{\partial t^2}, \quad \frac{\partial^2 I(x,t)}{\partial x^2} = v_{\text{signal}}^2 \cdot \frac{\partial^2 I(x,t)}{\partial t^2},$$

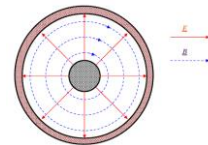
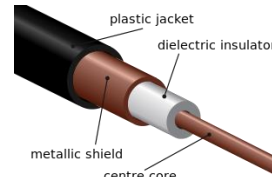
where v_{signal} is the propagation speed of the EM wave in the coaxial cable:

$$v_{\text{signal}} = \frac{1}{\sqrt{\tilde{L}\tilde{C}}}$$

the signal travels at approximately $\sim 2/3$ the speed of light in coaxial cables



D and d are the diameters of the outer and inner wires



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